

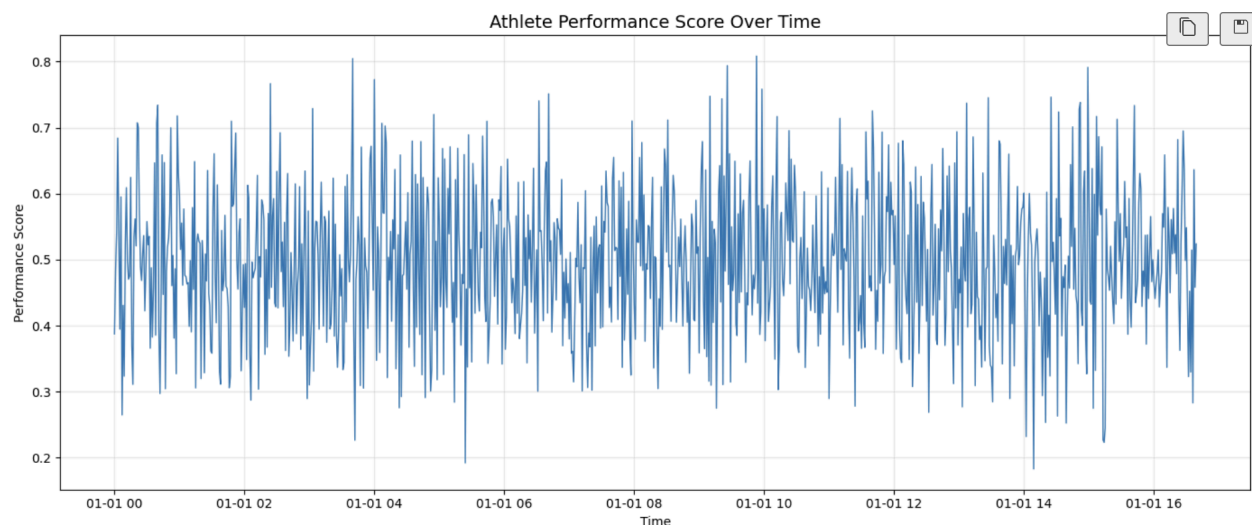
## Part 3 — Time Series

### Introduction

This report presents a time series forecasting analysis of athlete performance data. What: We analyzed 1000 observations of athlete physiological metrics collected over 16 hours, aiming to predict a continuous performance score. Why: Understanding how to forecast athlete performance using both classical and machine learning methods is valuable for real-time performance monitoring in sports. How: We implemented two approaches — Holt-Winters Exponential Smoothing as a classical method and Random Forest with engineered lag features as an ML method — and evaluated both using time series cross-validation and MAE/RMSE metrics.

### Step 1 — Data Preparation & Exploration

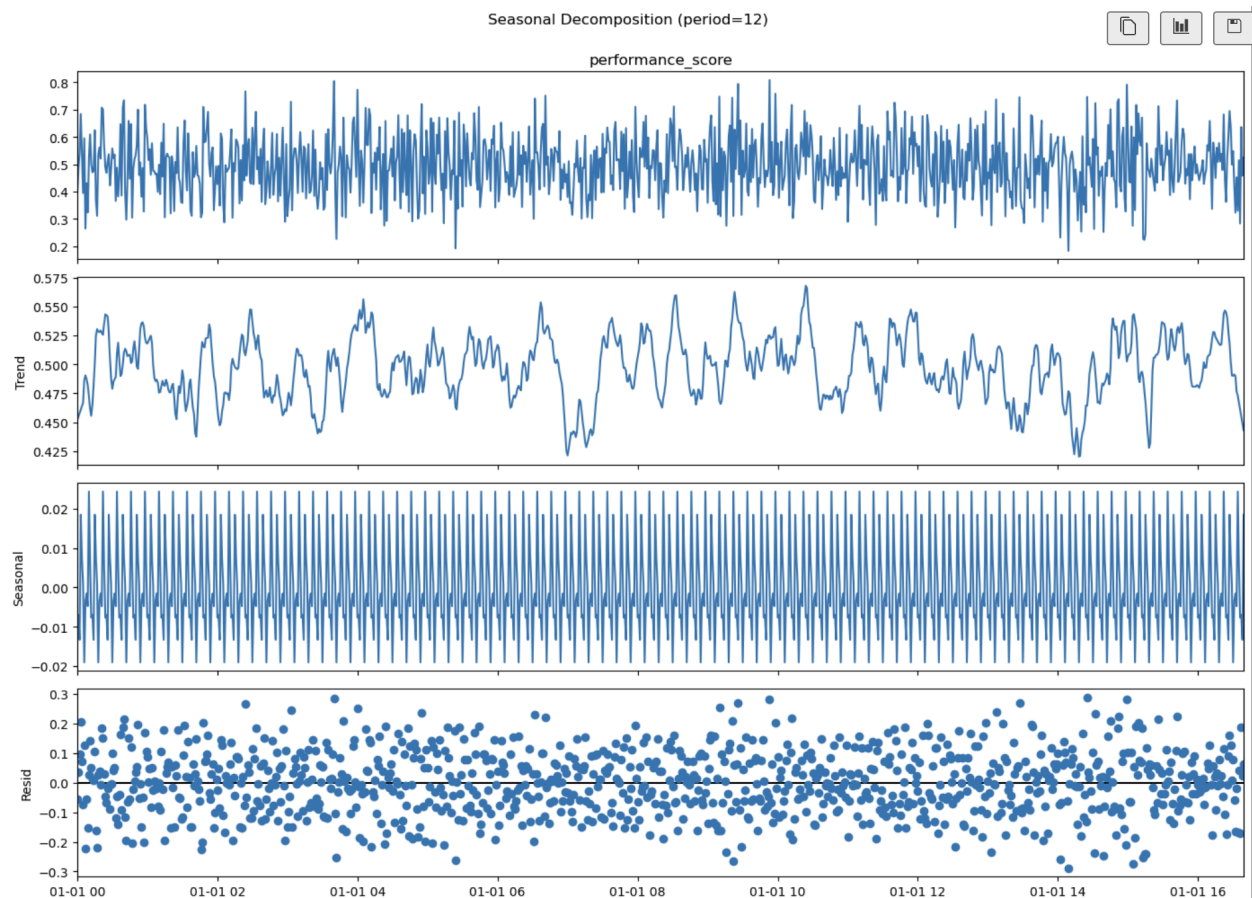
We worked with an athlete performance dataset containing 1000 time-ordered observations from 2023-01-01 00:00 to 16:39. The target, `performance_score`, is a continuous value mostly between 0.18 and 0.81 with an average around 0.50. The time series plot shows frequent short-term fluctuations around this level, with no clear long-term trend or strong repeating seasonal pattern in this 16-hour window.



## Step 2 – Stationarity Test

We used the Augmented Dickey–Fuller test to check stationarity of `performance_score`. The ADF statistic (about  $-21.05$ ) and a p-value near 0 are well below standard significance levels, so we reject the presence of a unit root. This indicates the series is stationary, and we do not need to differentiate it before modelling.

**Step 3 – Seasonality Detection.** We used seasonal decomposition and ACF/PACF plots to check for patterns in the data. The decomposition showed no meaningful seasonal component, and the ACF/PACF plots revealed no significant autocorrelation beyond lag 0. This indicates the series behaves mostly like white noise with no clear trend or seasonality.



### **Steps 4–5 – Data Split and Classical Forecasting**

We split the data chronologically into 80% for training (800 observations) and 20% for testing (200 observations), keeping the temporal order intact to avoid data leakage. For the classical forecasting method, we used Holt-Winters Exponential Smoothing with an additive trend and no seasonal component, since our earlier analysis showed the data does not have meaningful seasonality. The model was trained on the training set and used to forecast the 200 test points. It achieved an MAE of 0.0904 and an RMSE of 0.1164, which means predictions were on average about 0.09 away from the actual values — a reasonable result given that the performance scores range between 0.18 and 0.81.

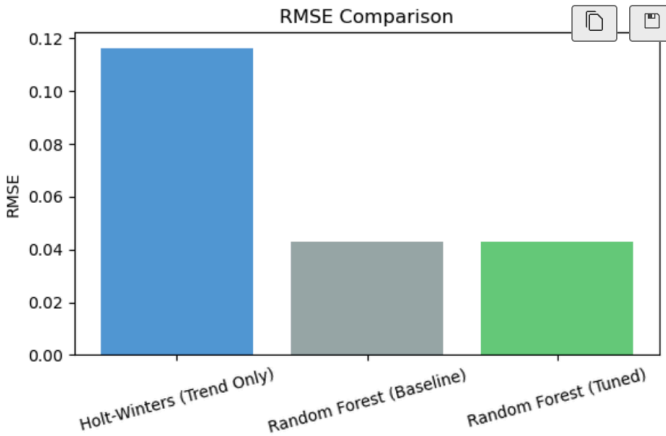
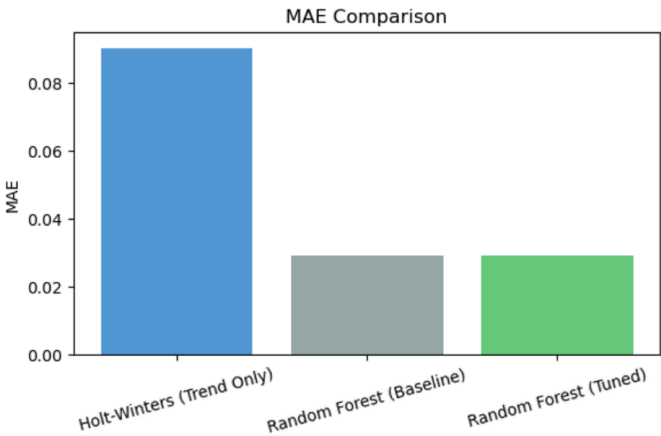
### **Steps 6–8 – ML Model with Lag Features**

To build our machine learning forecasting model, we first created lag features (`lag_1`, `lag_2`, `lag_3`, `lag_7`) and rolling statistics (rolling mean and standard deviation) from the `performance_score` series. This transformed the time series into a supervised learning problem with 7 input features. We then split the lagged data chronologically into 80% training (794 rows) and 20% testing (199 rows). We trained a Random Forest Regressor using a Pipeline that includes `StandardScaler`, as required by the project guidelines. The baseline model achieved an MAE of 0.0294 and an RMSE of 0.0428, which is significantly better than the Holt-Winters model, showing that the lag features capture useful short-term patterns in the data.

**Steps 9–10 – Hyperparameter Tuning** We used `GridSearchCV` with time-aware cross-validation to find the best Random Forest settings, testing 108 combinations on the training data only. The best parameters were 300 trees with no depth limit. The tuned model achieved an MAE of 0.0294 and an RMSE of 0.0428, which was very close to the baseline performance. This shows that the default Random Forest settings were already well-suited for our data.

**Steps 11–13 – Model Comparison and Visualizations.** We compared all three models using a results table and bar charts for MAE and RMSE. The Random Forest models significantly outperformed the Holt-Winters model, with MAE values around 0.029 compared to 0.090 for Holt-Winters. A forecast plot showed that both Random Forest predictions tracked the actual values much more closely than the Holt-Winters forecast. These visualizations clearly demonstrate that the ML approach with lag features is better suited for our data than the classical method.

|   | Method                    | MAE      | RMSE     |
|---|---------------------------|----------|----------|
| 0 | Holt-Winters (Trend Only) | 0.090363 | 0.116363 |
| 1 | Random Forest (Baseline)  | 0.029412 | 0.042773 |
| 2 | Random Forest (Tuned)     | 0.029406 | 0.042796 |



## Conclusion

For our athlete performance forecasting task, we implemented two approaches: a classical method (Holt-Winters) and a machine learning method (Random Forest with lag features). After testing for stationarity and seasonality, we found the data to be stationary with no meaningful seasonal pattern, so we used a trend-only Holt-Winters model. The Random Forest model achieved much better performance (MAE 0.0294 vs 0.0904), making it our recommended model for this task. The main trade-off is that Random Forest requires more feature engineering (lag and rolling features) and is less interpretable than Holt-Winters, but it captures the short-term patterns in the data more effectively. For real-time athlete monitoring, the Random Forest model would provide more accurate performance predictions.

